Creating Your Own Profile Website

A Beginner-Friendly HTML Lesson for IBSc Students
Designed for Teachers to Deliver with Practical Labs

HTML structures 100% of the web using tags

The Definition

- HTML stands for HyperText Markup Language.
- It is the standard language used to define the layout of web pages.
- Browsers read HTML files and turn them into the visible pages you see.

The Skeleton

- HTML forms the skeletal structure of every webpage.
- It specifies what is a heading, paragraph, image, or link.
- Just like a house framework, it defines where the walls and windows go.

How It Thinks

- HTML doesn't handle colors or logic.
- It uses simple text files with a suffix of .html
- It is incredibly robust—even if you make errors, the browser tries its best to display something.

Three core languages combine to build modern websites

HTML — Structure

- Role: The Bones
- Defines what text, images, and boxes appear on the page.
- Example Code:

```
<h1>My Profile</h1>
```

CSS — Styling

- Role: The Skin & Clothes
- Adds colors, layouts, backgrounds, fonts, and responsiveness.
- Example Code:

```
h1 { color: blue; }
```

JavaScript — Logic

- Role: The Muscles & Brain
- Handles clicks, pop-ups, calculations, animations, and APIs.
- Example Code:

```
alert('Hello World!');
```

VS Code installs in 3 steps to establish your workspace

1. Download

- Open your web browser.
- Visit: code.visualstudio.com
- Click the download button for your OS (Windows, macOS, or Linux).

2. Install

- Locate the downloaded installer in your Downloads folder.
- Double-click to run.
- Agree to terms and keep all default boxes checked.

3. Launch

- Search for 'Visual Studio Code' and launch the app.
- Welcome to your new home for coding!
- It starts with a clean, customizable dark theme.

Organizing files in 1 folder prevents broken image paths

Setup Instructions

1. Go to your Desktop or Documents.
2. Right-click -> New Folder.
3. Name the folder exactly: 'my-profile'

4. Open VS Code.
5. Click File -> Open Folder (or Cmd+O on Mac).
6. Choose the 'my-profile' folder you created.

Why this matters: Web projects MUST keep all assets under a single root folder so the browser can locate them.

Visual Workspace Layout

```
📁 my-profile-website/  (Root Folder)
│
├── 📄 index.html      (Your Homepage)
│
├── 📁 images/          (Folder for graphics)
│   └── 📷 photo.jpg   (Your profile image)
│
└── 📄 style.css       (Optional styling file)
```

This clear structure prevents 'file not found' errors when linking items!

Naming your file index.html creates the default homepage

How to Create It in VS Code

1. Hover over the 'my-profile' folder name in the VS Code sidebar.
2. Click the 'New File' icon (looks like a page with a +).
3. Type the filename exactly: index.html
4. Press Enter to create the file.

Critical Rules:

• Must be entirely lowercase (index, not Index).

• Must end with .html extension.

• Do not use spaces!

The Special Status of index.html

• When a browser visits a web address (e.g., google.com), the server automatically searches for 'index.html' inside that folder to serve as the default homepage.

• If you don't name your file index.html, servers won't know which file to load first, resulting in a 'Directory Listing' or an 'Error 404: Page Not Found'!

Five essential elements form the HTML document skeleton

index.html Boilerplate

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>My Profile Website</title>
</head>
<body>
    <!-- Your visible website content goes here! -->
    <h1>Hello World!</h1>
</body>
</html>
```

The 5 Skeleton Pillars

1. <!DOCTYPE html>: Informs the browser that this is a modern HTML5 document.
2. <html>: The outer container of the entire web document.
3. <head>: Contains invisible settings, character sets, and the tab title.
4. <title>: Defines the page title shown on the browser tab.
5. <body>: Contains the visible content of your webpage (headings, images, text).

Most HTML tags require opening and closing pairs

Anatomy of an HTML Element



Key Tag Guidelines:

- Closing tags are identical to opening tags but include a forward slash (/).
- Tags must nest properly: text, NOT text

Six heading levels and paragraphs organize text content

HTML Code

```
<h1>Alex's Portfolio</h1>

<h2>About Me</h2>
<p>I am a first-year science student at IBSc,
exploring web design.</p>

<h3>My Hobbies</h3>
<p>Coding, playing guitar, and photography.</p>
```

Browser Preview

Alex's Portfolio (Very Large & Bold)

About Me (Medium & Bold)

I am a first-year science student at IBSc, exploring web design.

My Hobbies (Small & Bold)

Coding, playing guitar, and photography.

Unordered and ordered lists structure items in 2 ways

Unordered (Bullets):

```
<!-- Bullet list of skills -->
<ul>
    <li>HTML Coding</li>
    <li>VS Code Basics</li>
    <li>File Management</li>
</ul>

Browser Output:
  • HTML Coding
  • VS Code Basics
  • File Management
```

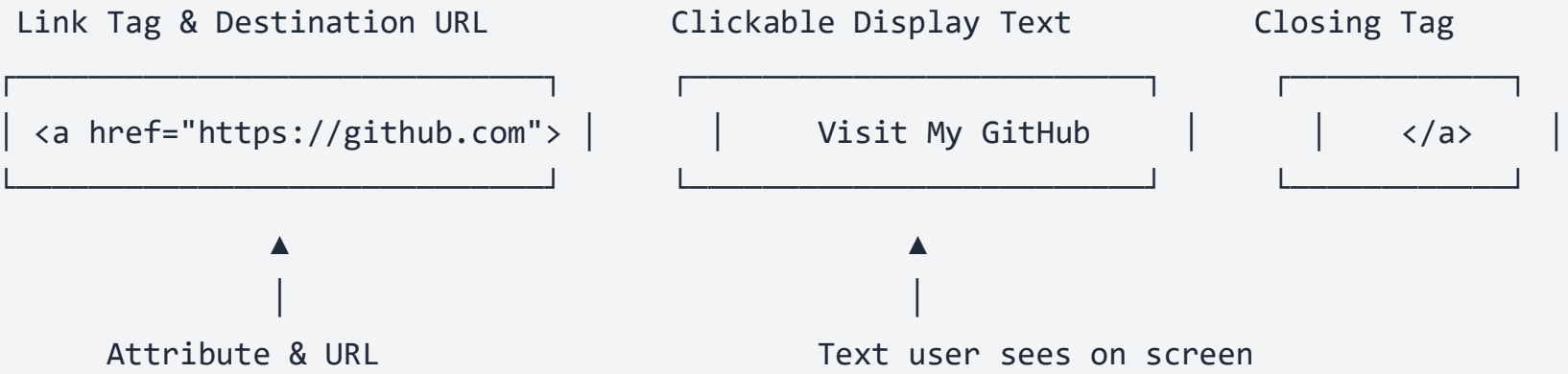
Ordered (Numbers):

```
<!-- Numbered steps to create files -->
<ol>
    <li>Create folder</li>
    <li>Open in VS Code</li>
    <li>Create index.html</li>
</ol>

Browser Output:
  1. Create folder
  2. Open in VS Code
  3. Create index.html
```

The anchor tag links your site to any URL on the web

Anatomy of a Hyperlink



Key Link Attributes:

- href: The destination address of the website you want to link to (must include http/https).
- target="_blank": (Optional) Tells the browser to open the link in a new tab!

Image tags require 2 key attributes: src and alt text

The Image Tag Structure

```
<!-- Simple self-closing image tag -->
<img src="images/avatar.jpg" alt="Alex Profile
Photo">

Anatomy:
• src: Path to your image file.
• alt: Description of the image.
• Note: No closing tag needed!
```

Critical Guidelines

• src (Source): Tells the browser where the image lives. It can be a local path (images/photo.jpg) or a web URL.

• alt (Alternative Text):

- Crucial for blind/visually-impaired users using screen readers.
- Displays on the screen if the image fails to load.
- Heavily improves search engine rankings (SEO).

Double-clicking your file opens it directly in any browser

Steps to View Your Page

1. Open your computer's File Explorer (Windows) or Finder (Mac).

2. Navigate into your 'my-profile' folder.

3. Double-click the index.html file.

4. Your default web browser (Chrome, Edge, Safari) will open and display your website!

⚠ The 'No Live Reload' Limitation

• The address bar will show: file:///C:/Users/... index.html

• Problem: If you make changes in VS Code, they won't automatically show up in the browser.

• Manual Fix: You must SAVE your file in VS Code (Ctrl+S) and then click REFRESH (F5) in the browser.

Let's look at a much better way to work! 👇

Live Server automatically refreshes your browser on every save

1. Install Extension

• Click the Extensions icon on the left sidebar in VS Code (looks like a square panel).

• Search for: Live Server

• Look for Ritwick Dey's popular extension and click Install.

2. Start Server

• Open index.html in the editor.

• Click the Go Live button in the bottom-right status bar.

• Alternatively: Right-click index.html in sidebar and select 'Open with Live Server'.

3. Real-Time Coding

• Live Server spins up a mini local web server on your computer (127.0.0.1:5500).

• Any time you edit your HTML and SAVE, the browser reloads automatically!

• Instant feedback!

Three common HTML mistakes can break your page layout

1. Unclosed Tags

```
<!-- Broken Code -->
<p>I love coding.
<h2>About Me</h2>
```

Why it breaks:
The browser thinks the heading is part of the paragraph, combining fonts and ruining styles.

2. Missing Quotes

```
<!-- Broken Code -->
<a href=https://google.com>
```

Why it breaks:
If there are other attributes, the browser will read them all as one single, broken URL.

3. Typo in Tags

```
<!-- Broken Code -->
<h11>My Profile</h11>
```

Why it breaks:
Web browsers won't recognize '<h11>' and will display it as plain, small body text.

The 4-step debugging loop helps you solve code errors quickly

1. Read

Observe the issue.

What exactly is wrong in the browser? Is an image missing? Is heading text too small? Is a link unclickable?

2. Find

Locate the code.

Open index.html in VS Code. Go to the line where that content is written. Inspect tags and spelling.

3. Fix

Make the edit.

Correct the typo, add the missing closing tag, or insert the forgotten quote marks. Save (Ctrl+S).

4. Test

Verify the result.

Look back at your browser (refreshed automatically by Live Server). Did the fix work? If not, loop back!

The W3C Validator scans your HTML code to find syntax mistakes

What is the W3C Validator?

• The W3C (World Wide Web Consortium) is the official governing organization that writes the specifications for HTML.

• They provide a free validator at validator.w3.org.

• It scans your HTML code and points out syntax errors, mismatched tags, and obsolete rules with absolute precision.

How to Validate Your Code

1. Copy all the code inside your index.html file (Ctrl+A -> Ctrl+C).

2. Go to validator.w3.org.

3. Select the 'Validate by Direct Input' tab.

4. Paste your code into the text area.

5. Click 'Check'. Read any red error logs, fix them in VS Code, and retest!

Create and submit your own profile website using these 5 skills

Your Lab Project: My Profile Website

Build a personal profile homepage containing:

• Main Title: 'My Profile' using <h1>.

• About Me: 1-2 paragraph introduction using <p>.

• Profile Image: <img> with proper 'src' and 'alt'.

• My Skills: An unordered bullet list (<ul> +) showing at least 3 skills or hobbies.

• Social Link: A working link (<a> + href) taking users to your GitHub, LinkedIn, or Email.

Challenge: Add <h2> subtitles to organize sections.

Submission Requirements

1. Code Quality Check:

• Paste your HTML into validator.w3.org.

• Resolve any errors until you get the green 'No errors' message.

2. Packaging Your Project:

• Zip your entire 'my-profile' project folder.

• Ensure your images folder and index.html are inside the zip.

3. Submit the .zip file to the IBSc class portal!